

Bill of Materials (BOM)

A Bill of Materials (BOM) was created to identify all components required for the assembly of the desktop PCB pick-and-place machine. The BOM includes information such as vendor, part name or number, unit cost, quantity, and total cost for each component.

Parts were sourced from suppliers including **McMaster-Carr, Digi-Key, Amazon, and AliExpress**, along with several **custom-manufactured aluminum components** used for structural and mounting elements. The components were organized by subsystem, including the **reel feeder, Z-axis motion mechanism, camera module, main frame, and base structure**.

The total cost of the prototype system was estimated by summing the cost of all listed components, resulting in an approximate total cost of **\$5,404.89 USD**.

Assembly	Vendor	Part Name / Number 	Cost/Unit [USD]	Qty	Total Cost [USD]
1. Reels Feeder	Digi-Key	EE-SX3070; Optical sensor	\$7.26	15	\$108.90
	McMaster Carr	624 Plastic Ball Bearing	\$7.48	30	\$224.40
	McMaster Carr	Lightweight Sprocket for ANSI 25 Chain, 25 Teeth, for 1/2" Shaft Diameter	\$32.88	15	\$493.20
	Custom made	NEMA 23 worm gearbox, Modified*	\$6.55	15	\$98.25
2. Z Motion Mechanism	McMaster Carr	Stepper Motor NEMA 14, 7.5 in.-oz. Maximum Holding Torque	\$77.36	1	\$77.36
	McMaster Carr	Stepper Motor NEMA 8, 2.8 in.-oz. Maximum Holding Torque	\$115.14	2	\$230.28
	McMaster Carr	Gear Rack	\$20.00	2	\$40.00
	McMaster Carr	Corrosion-Resistant Ball Bearing Carriage for 7 mm Wide Rail	\$75.92	2	\$151.84

	McMaster Carr	7 mm Wide 440 Stainless Steel Guide Rail for Ball Bearing Carriage	\$28.80	2	\$57.60
	McMaster Carr	Steel Hex Nut Medium-Strength, Class 8, M2 x 0.4 mm Thread	\$0.04	8	\$4.00
	McMaster Carr	Black-Oxide Alloy Steel Hex Drive Flat Head Screw 90 Degree Countersink, M2 x 0.40 mm Thread, 12 mm Long	\$0.36	8	\$8.93
	McMaster Carr	Thread-Forming Screws for Soft Metal Zinc-Plated Steel, Torx-Plus Drive, M1.4 Thread, 4 mm Long	\$0.52	8	\$12.98
	McMaster Carr	Anodized Multipurpose 6061 Aluminum Sheet 1/4" Thick	\$96.25	1	\$96.25
	McMaster Carr	Metal Gear - 20 Degree Pressure Angle Round with M4 Set Screw, 0.5 Module, 24 Teeth	\$15.87	1	\$15.87
3. Camera	e-con Systems	e-CAM55_CUMI0521_MOD - 5MP AR0521 MIPI Camera Module	\$79.00	2	\$158.00
4. Main Frame	McMaster Carr	T-slotted framing 1 / 47065T411	\$4.48	4	\$17.92
	McMaster Carr	T-slotted framing 2 / 47065T411	\$8.97	2	\$17.94
	McMaster Carr	T-slotted framing for rail attachment / 47065T418	\$6.34	4	\$25.36
	McMaster Carr	The rail attachment / 1748N11	\$16.53	4	\$66.12
	McMaster Carr	500mm rail / 6709K33	\$160.00	2	\$320.00
	McMaster Carr	Silver Corner Bracket / 47065T239	\$8.00	9	\$72.00

	McMaster Carr	T-Slotted Framing Fasteners / 47065T147	\$6.12	24	\$146.88
	McMaster Carr	Alloy Steel Socket Head Screw / 91290A150	\$13.38	20	\$267.60
	McMaster Carr	T-Slotted framing 3 / 47065T418	\$6.34	2	\$12.68
	McMaster Carr	Belt for the x-y motion / 6484K504	\$8.49	3	\$25.47
	McMaster Carr	Timing Belt Pulley / 1277N11	\$12.36	6	\$74.16
	McMaster Carr	Ultra Low Profile Socket Head Screw / 90358A009	\$4.89	24	\$117.36
	McMaster Carr	Steel Thin Hex Nut / 90695A035	\$0.03	24	\$0.66
	custom made	Plates for horizontal motor / Custom made (Aluminum 6061)	\$3.66	4	\$14.64
	custom made	NEMA 17 motor bracket / "Custom made (Aluminum 6061)"	\$1.40	5	\$7.00
	McMaster Carr	Ball bearing / 60355K505	\$6.75	3	\$20.25
	McMaster Carr	Steel Thin Hex Nut / 90370A204	\$0.07	28	\$2.00
	McMaster Carr	Ultra Low Profile Socket Head Screw / 90358A141	\$8.05	10	\$80.50
	McMaster Carr	Steel Thin Hex Nut / 90695A152	\$0.10	10	\$1.01
	McMaster Carr	Ultra Low Profile Socket Head Screw / 90695A152	\$4.29	24	\$102.96
	custom made	Plates for horizontal motor / Custom made (Aluminum 6061)	\$3.66	1	\$3.66
	McMaster Carr	NEMA 17 Stepper motor / 6627T92	\$69.72	1	\$69.72
	custom made	Timing Belt holder / Custom made (Aluminum 6061)	\$0.25	12	\$3.00

	custom made	Plate for bearing carrier and frame left / "Custom made (Aluminum 6061)"	\$4.80	1	\$4.80
	McMaster Carr	Ball Bearing Carriage / 6709K12	\$118.97	2	\$237.94
	McMaster Carr	Ball bearing / 7208K51	\$40.06	1	\$40.06
	McMaster Carr	High strength class 10.9 steel hex head screw / 91310A530	\$0.28	2	\$0.56
	McMaster Carr	Medium strength steel hex nut / 95505A711	\$4.90	2	\$9.80
	McMaster Carr	Rotary shaft / 1346K11	\$9.14	1	\$9.14
	McMaster Carr	T-slotted Framing Brackets/ 3136N173	\$13.57	4	\$54.28
	McMaster Carr	Ultra Low Profile Socket Head Screw / 90358A129	\$4.70	12	\$56.40
	custom made	Plate for bearing carrier and frame right / Custom made (Aluminum 6061)	\$5.95	1	\$5.95
	McMaster Carr	Stepper Motor / 6627T65	\$71.91	1	\$71.91
	McMaster Carr	T-slotted framing / 47065T10	\$12.17	1	\$12.17
	McMaster Carr	Track Roller Carriage for T-Slotted Framing / 9904K1	\$84.47	2	\$168.94
	custom made	Plate for bearing carriers for the middle motion / custom made(aluminum 6061)	\$4.28	1	\$4.28
	McMaster Carr	Alloy Steel Low Profile Socket Head Screw / 93070A143	\$0.25	4	\$0.99
	custom made	Bearing holder / Custom made (PP)	\$0.01	3	\$0.03
	McMaster Carr	Rod for Spacers / 1346K511	\$12.26	1	\$12.26

	McMaster Carr	Ultra Low Profile Socket Head Screw / 90358A118	\$4.30	2	\$8.60
	McMaster Carr	Steel Thin Hex Nut / 90370A203	\$0.06	2	\$0.11
5. Base Structure	McMaster Carr	8-32 Flathead 3/8 in. Bolts / 91253A192	\$0.12	12	\$12.21
	McMaster Carr	6-32 Low Profile 3/8 in. Bolts / 93615A215	\$0.30	9	\$7.39
	McMaster Carr	10-24 Flanged Button Head 1/2 in. Screws / 97654A142	\$0.50	16	\$12.38
	McMaster Carr	M5 x 0.8 Low Profile 5 mm Bolts / 92855A501	\$0.96	4	\$9.60
	McMaster Carr	M5 x 0.8 Low Profile 10 mm Bolts / 92855A510	\$0.24	10	\$12.10
	Aliexpress	Screw Actuated Linear Slide / FUYU FSK30J	\$121.43	3	\$364.29
	McMaster Carr	T-Slot Framing Structural Brackets (L Shaped) / 47065T239	\$8.00	8	\$64.00
	Amazon	36mm LED Lights / Aramox Aramoxf0hx6mvu9g	\$0.80	2	\$5.92
	McMaster Carr Custom Made	Base Box Bottom and Longer Side Walls [Raw Material: Low Carbon Steel Plate 3/8 in. Thick] / 1388K281	\$370.82	40x50 cm (15.74x19.68 in) 19.68x2.36 in * 2	\$370.82
	McMaster Carr Custom Made	Base Lid and Shorter Side Walls[Raw Material: Low Carbon Steel Plate 3/4 in. Thick] / 1388K453	\$560.57	15.74x2.36 in * 2 40x50 cm (15.74x19.68 in)	\$560.57
	McMaster Carr Custom Made	Low Carbon Steel Bar 2 wide x 1/2 in. Thick / 8910K949	\$18.70	10.5x3.8x1 cm (~4x2x1/2 in) * 2	\$18.70
	McMaster Carr Custom Made	Low Carbon Steel Bar 1 wide x 1 in. thick / 9143K21	\$21.94	10.5x2x2 cm (~4x1x1 in) * 2	\$21.94
TOTAL COST [USD]:					\$5,404.89